

Tianhao Wu

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[Sm0kyWu.github.io/CV/CV.html](https://github.com/Sm0kyWu)



Google Scholar

Education Background

Beihang University (BUAA)

Beijing, China

Bachelor of Engineering in Computer Science and Technology

Sept 2017 - Jun 2021

- **Honors:** Merit Student of the School of Computer Science, Excellent Student of University.

Nanyang Technological University (NTU)

Singapore, Singapore

Ph.D. Candidate, S-Lab, Computer Science and Engineering. Supervisor: Prof. Tat-Jen Cham

Jan 2022 - Present

- **Research focus:** Computer Vision - 2D and 3D generation, reconstruction and editing.

Internship

Shanghai Lin-Gang Sense Auto Intelligent Technology Co., Ltd., Beijing Branch (SenseTime Inc.)

Beijing, China

Full-time research intern in driving planning group

Oct 2021 - Dec 2021

- Shanghai Lin-Gang Sense Auto Intelligent Technology Co., Ltd., Beijing Branch is the Beijing branch office of Shanghai Lin-Gang Sense Auto Intelligent Technology Co., Ltd. (formerly Shanghai SenseTime Lingang Intelligent Technology Co., Ltd.), which is a 100% wholly-owned subsidiary of SenseTime Group Inc.
- Organizing the code of the autopilot planning part.
- Research the latest planning algorithms and apply them to autonomous driving system.

Beijing Youzhuju Network Technology Co., Ltd. (ByteDance Inc.)

Beijing, China

Full-time technical intern in AI-Lab intelligent voice group

Feb 2021 - May 2021

- Beijing Youzhuju Network Technology Co., Ltd. is a wholly-owned subsidiary of ByteDance (Hong Kong) Ltd., serving as the legal entity for ByteDance's Beijing operations.
- Mainly responsible for voice activity detection(VAD).
- Build VAD datasets, train and fine-tune models according to different requirements. Integrate the latest research.
- On the business side, responsible for VAD model requirements including education and multi-language service.

Research and Publications

Amodal3R: Amodal 3D Reconstruction from Occluded 2D Images (ICCV 2025)

Jun 2024 - Mar 2025

- Given partially visible objects within images, our method reconstructs semantically meaningful 3D assets with reasonable geometry and plausible appearance, significantly outperforming current baselines.
- This work is done under the collaboration and guidance of Prof. Andrea Vedaldi and Dr. Chuanxia Zheng, Visual Geometry Group, University of Oxford.

ClusteringSDF: Self-organized neural implicit surfaces for 3D decomposition (ECCV 2024)

Jul 2023 - Mar 2024

- Proposed a novel segmentation method for 3d indoor scenes via object-compositional SDF.
- Our method doesn't require ground-truth segment labels for supervision, instead it can take inconsistent labels of multiple camera views from 2D pre-trained segmentation models.
- The segmentation accuracy outperforms current SOTA 3D semantic/instance segmentation methods.

LTOS: Layout-controllable Text-object Synthesis (ICASSP 2025, coauthor)

Sept 2023 - Apr 2024

- Proposed a framework that generates images with clear, legible visual text and plausible objects.
- Constructed a visual-text rendering module to synthesize text and employ an object-layout control module to generate objects while integrating the two modules to harmoniously generate text content and objects in images.
- Our method outperforms the state-of-the-art in both text rendering and layout-to-image tasks.

PanoDiffusion: 360-degree panorama outpainting via diffusion (ICLR 2024)

Jan 2023 - Jun 2023

- Proposed a new bi-modal latent diffusion structure that utilizes both RGB and depth panoramic data to better learn spatial layouts and patterns during training, works surprisingly well to outpaint normal depth-free RGB images .
- Designed a novel technique of introducing progressive camera rotations during each diffusion denoising step, which leads to substantial improvement in achieving panorama wraparound consistency.

Predictive State Representation

Jan 2022 - Oct 2022

- Aimed to implement a competitive model-based model using the concept of predictive state representation.
- Tested our model in both continuous and discrete environments, including the Mujoco and Atari games.
- In this process, introduced information bottlenecks and VAE to allow the model to achieve better results.

Image Segmentation of Breast Tumors

Jul 2020 - Jun 2021

- Developed a CNN model to automatically annotate the region of tumors in key frames of ultrasonoscopy (US) and ultrasonic videos (CEUS) of the breast, improving doctors' efficiency in the treatment.
- Utilized the segmentation of US to assist the division of tumor regions in the CEUS.

Competitions

30th Science & Technology Competition at Beihang University (Second price)

Jan 2020 - May 2020

Alert-situation Analysis and Alarm-Receiving Platform Based on Text-mining

- Developed a functional platform to automatically analyze cases and assist in case-solving.
- Realized the interaction between the front-end and back-end by delivering the typed information from the front-end to the analyzing interface to extract and classify keywords.

2019 CCF Computing Intelligence Contest (CCF BDCI)

Sept 2019 - Nov 2019

Quality Analysis of Discrete Work-pieces

- Designed a model to predict the quality of unknown work-pieces based on the known parts.
- Analyzed statistics with different machine learning models, and obtained the optimal results with fusion model.
- Ranked top 3% in over 2000 teams.

Engineering Project

Development of An Embedded Robot

Mar 2020 - Jun 2020

- Build an embedded robot with multiple functions, including voice control and capturing with manipulators.
- Realized voice control and active control based on controlling the movement of robots with different signals.

Data Analysis for Didi Chuxing

Nov 2019 - Dec 2019

- Studied data of Didi Chuxing in Haikou City and analyzed the travel experiences of users to predict future travels.
- Conducted k-means clustering analysis on departure spots and destinations individually and classified the two types of data to study the possibility of journeys happening from one departure spot to one destinations.
- Applied bigdata methods to calculate the possibility of journeys respectively to study the travel modes.

Telegram Recommendation Robot

Jul 2019 - Sept 2019

Supervisor: Fan Zhang from MIT

- Developed a conversational robot that could recognize users' intention and offer real-life recommendations.
- Added the registration function and verified the identity of users with SQL database and encryption.
- Enabled the software to recognize users' intentions with NLP.

Skills

Programming Python, LATEX, C/C++, R, Matlab...

Language Mandarin Chinese, English

Academic Services

Conference Reviewer CVPR, ICCV, ICLR, ECCV, ACMMM, 3DMV

Journal Reviewer TMM